

MongoDB CRUD Operations with Update, Unset, Rename, Aggregate and Delete Functions

```
import org.bson.Document;

import com.mongodb.client.AggregateIterable;
import com.mongodb.client.FindIterable;
import com.mongodb.client.MongoClient;
import com.mongodb.client.MongoClients;
import com.mongodb.client.MongoCollection;
import com.mongodb.client.MongoDatabase;
import com.mongodb.client.model.Filters;
import com.mongodb.client.model.Updates;

public class MongoDBConnection {

    static MongoCollection<Document> collection;

    // CREATE

    public static void createData(int id, String name,
                                   String department, double cgpa) {

        Document student = new Document("id", id)
            .append("name", name)
            .append("department", department)
            .append("cgpa", cgpa);

        collection.insertOne(student);
```

```
    System.out.println("Document Inserted Successfully : " + name);  
}
```

```
// READ
```

```
public static void readData() {
```

```
    System.out.println("\nStudent Records:");
```

```
    FindIterable<Document> students = collection.find();
```

```
    for (Document doc : students) {
```

```
        System.out.println(doc.toJson());
```

```
    }
```

```
}
```

```
// UPDATE (SET)
```

```
public static void updateCGPA(int id, double newCgpa) {
```

```
    collection.updateOne(  
        Filters.eq("id", id),
```

```
        Updates.set("cgpa", newCgpa));
```

```
    System.out.println("\nCGPA Updated Successfully for ID : " + id);
```

```
}
```

```
// UNSET
```

```
public static void removeDepartment(int id) {
```

```
collection.updateOne(
    Filters.eq("id", id),
    Updates.unset("department"));

System.out.println("\nDepartment Field Removed for ID : " + id);
}
```

// RENAME

```
public static void renameField(int id) {
```

```
collection.updateOne(
    Filters.eq("id", id),
    Updates.rename("name", "studentName"));
```

```
System.out.println("\nField Renamed Successfully for ID : " + id);
}
```

// DISPLAY SPECIFIC RECORD

```
public static void displayRecord(int id) {
```

```
Document student =
    collection.find(Filters.eq("id", id))
        .first();
```

```
System.out.println("\nStudent Record:");
```

```
if (student != null) {
```

```
        System.out.println(student.toJson());  
    }  
}
```

```
// AGGREGATE (AVERAGE CGPA)
```

```
public static void averageCGPA() {
```

```
    System.out.println("\nAverage CGPA:");
```

```
    AggregateIterable<Document> result =
```

```
        collection.aggregate(java.util.Arrays.asList(  
            new Document("$group",
```

```
                new Document("_id", null)
```

```
                    .append("averageCGPA",
```

```
                        new Document("$avg", "$cgpa"))
```

```
        ));
```

```
    for (Document doc : result) {
```

```
        System.out.println(doc.toJson());
```

```
    }
```

```
}
```

```
// DELETE
```

```
public static void deleteData(int id) {
```

```
    collection.deleteOne(Filters.eq("id", id));
```

```
    System.out.println("\nDocument Deleted Successfully for ID : " + id);
```

```
}
```

```
public static void main(String[] args) {
```

```
    try (MongoClient mongoClient =
```

```
        MongoClient.create("mongodb://localhost:27017")) {
```

```
        // Database
```

```
        MongoDB database =
```

```
            mongoClient.getDatabase("collegedb");
```

```
        // Collection
```

```
        collection =
```

```
            database.getCollection("Students");
```

```
        // Clear old records
```

```
        collection.deleteMany(new Document());
```

```
        // CREATE
```

```
        createData(101, "John", "CSE", 8.7);
```

```
        createData(102, "Anushka", "IT", 9.5);
```

```
        createData(103, "Manas", "ECE", 8.2);
```

```
        createData(104, "Rahul", "CSE", 7.9);
```

```
        // READ
```

```
        readData();
```

```
        // UPDATE
```

```
updateCGPA(101, 9.2);

// UNSET
removeDepartment(101);

// RENAME
renameField(101);

// DISPLAY UPDATED RECORD
displayRecord(101);

// AGGREGATE
averageCGPA();

// DELETE
deleteData(104);

System.out.println("\nAfter Deletion:");
readData();

} catch (Exception e) {
    e.printStackTrace();
}
}
```

workspace - Java - MONGODB/src/MongoDBConnection.java - Eclipse

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Package Explorer

- Mongo
- MONGODB
 - src
 - (default package)
 - MongoDBConnection.java
 - MongoMultipleRecords.java
 - JRE System Library [JavaSE-1.8]
 - Referenced Libraries
 - MYSQL_CRUD

MongoDBConnection.java

```
133
134 // READ
135 readData();
136
137 // UPDATE
138 updateCGPA(101, 9.2);
```

Task List

Outline

Problems Javadoc Declaration Console

<terminated> MongoDBConnection [Java Application] C:\Program Files\Java\jre1.8.0_491\bin\javaw.exe (05-Jun-2026, 8:17:39 PM)

Jun 05, 2026 8:17:39 PM com.mongodb.driver.logging.Loggers shouldUseSLF4J

WARNING: SLF4J not found on the classpath. Logging is disabled for the 'org.mongodb.driver' component

Document Inserted Successfully : John

Document Inserted Successfully : Anushka

Document Inserted Successfully : Manas

Document Inserted Successfully : Rahul

Student Records:

```
{ "_id": {"$oid": "6a22e18c34a1db61b081ee29"}, "id": 101, "name": "John", "department": "CSE", "cgpa": 8.7}
{"_id": {"$oid": "6a22e18c34a1db61b081ee2a"}, "id": 102, "name": "Anushka", "department": "IT", "cgpa": 9.5}
{"_id": {"$oid": "6a22e18c34a1db61b081ee2b"}, "id": 103, "name": "Manas", "department": "ECE", "cgpa": 8.2}
{"_id": {"$oid": "6a22e18c34a1db61b081ee2c"}, "id": 104, "name": "Rahul", "department": "CSE", "cgpa": 7.9}
```

CGPA Updated Successfully for ID : 101

Department Field Removed for ID : 101

Field Renamed Successfully for ID : 101

Student Record:

```
{ "_id": {"$oid": "6a22e18c34a1db61b081ee29"}, "id": 101, "cgpa": 9.2, "studentName": "John"}
```

Average CGPA:

```
{ "_id": null, "averageCGPA": 8.7}
```

Document Deleted Successfully for ID : 104

After Deletion:

Student Records:

```
{ "_id": {"$oid": "6a22e18c34a1db61b081ee29"}, "id": 101, "cgpa": 9.2, "studentName": "John" }
{"_id": {"$oid": "6a22e18c34a1db61b081ee2a"}, "id": 102, "name": "Anushka", "department": "IT", "cgpa": 9.5}
{"_id": {"$oid": "6a22e18c34a1db61b081ee2b"}, "id": 103, "name": "Manas", "department": "ECE", "cgpa": 8.2}
```

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